Ultimate Nutrition

Antioxidant

Antioxidants provide a layer of protection for the cells and tissues of the body. Specifically, antioxidants protect against free radical damage. People must breathe in oxygen to live. Oxygen is necessary for all essential bodily functions. However, a small amount of oxygen gets loose and produces unstable by-products called free radicals. Body processes, such as metabolism, as well as environmental factors, including pollution and cigarette smoke, can produce free radicals. An overload of free radicals in the body causes damage to the cells.

Strenuous exercise increases the body's production of free radicals, which, in turn, can cause muscle damage, which manifests as swollen or painful muscles. While exercise increases the body's natural defense against free radicals, athletes who are doing intense training may benefit from the addition of antioxidant supplements to their diets.

Antioxidants work in a variety of ways to reduce the effects of free radicals. They can greatly decrease the damage caused by free radicals, stop them from forming to begin with, or "oxidize" them by combining with them and neutralizing their harmful effects through stabilization.

Vitamin C, a water-soluble vitamin, is also known as ascorbic acid. Most of the vitamin C in the diet comes from fruits and vegetables. However, since vitamin C is water soluble, cooking can destroy the vitamin C in a food. Vitamin C is essential for the healing of wounds, and for the repair and maintenance of cartilage, bones, and teeth. Beyond that, vitamin C acts against the toxic effects of environmental pollutants by stimulating liver detoxifying enzymes and acts as an anti-inflammatory.

Zinc is an essential nutrient required by humans and animals alike. Zinc is a mineral that is vital to healthy living. Zinc is probably most widely known for its ability to prevent colds and shorten the duration of colds. This is because of zinc's powerful ability to strengthen the immune system and improve white blood cell count. It plays a variety of biological roles in the body including catalytic, structural, and regulatory roles.

Because vitamin E is found in oils, people who follow a low-fat diet may not get enough. Vitamin E is effective in preventing the oxidation of polyunsaturated fatty acids. Additionally, Vitamin E is helpful in the prevention of oxidation in the lungs, where strong oxidizing agents nitrogen dioxide and ozone, components of air pollution, are particularly harmful to people exercising. Vitamin E protects white and red blood cells, helping the body's immune system.

Beta-carotene is a member of the carotenoid family. Found mainly in plants, carotenoids provide the vibrant red, yellow, green, and orange colors of fruits and vegetables, with carrots being a major contributor of beta-carotene. Typically, beta-carotene is a conditionally essential nutrient, but when one's intake of vitamin A is low, beta-carotene becomes an essential nutrient, meaning that it must be obtained from food and cannot be manufactured by the body.

Cysteine is a sulfur-containing non-essential amino acid. It's required for healthy skin, detoxification of the body and the production of collagen (used for skin elasticity and texture). It is used by the body to manufacturer taurine and glutathione.

Glutathione is your body's most abundant natural antioxidant. Glutathione helps eliminate toxins from the body, and keeps the eyes, central nervous system, and immune system healthy and strong. Glutathione also helps turn carbohydrates into energy, and prevents the buildup of oxidized fats that may contribute to atherosclerosis.

Selenium is an essential trace mineral. The amount of selenium found in food is directly related to the amount of selenium in the soil in which the food was grown. It is necessary for healthy immune function and is tied to antibody production.

Antioxidants protect our cells from damage caused by radicals. Because of their function in cell protection, the benefits of antioxidants are far-reaching and not contained to a single aspect of our health.

Selected References:

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Frequently Asked Questions:

1. What are oxidants?

Oxidants are active oxygen components that are the cause of oxidative damage to biological molecules and can initiate various severe diseases.

2. What are some sources of oxidants?

Cigarette smoke, exercise, air pollutants, radiation and tissue inflammation are all possible sources of oxidants.

3. What are antioxidants?

Antioxidants act as a defense against oxidative damage by helping to inhibit oxidation reactions.